Instruction to Generate SPA & Degenerate ETPA Figures using Matlab code

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Instructions for using the Matlab codes to generate the SPA & ETPA combination figures
Note that the code for these figures are more complicated than for generally all of the other figures that I made during the summer.

The code for these figures also tends to need more modification due to removing the tail when the idler energy begins to enter resonance with the SPA.

Which can also change the range of the plots, which affects the placement of several of the various text that are on the plots, which have been manually positioned for consistency within the plot but if any of the values that specify the size of the plot change (both actual size and range of axes) they need to be repositioned.

Just a warning that these are less one size fits all than what the rest of what I did

1. Place all the provided Matlab code and the ETPA and SPA spectrum data in the same directory.
   - See SPA AND Degenerate_ETPA_figure OR the Fortran section of the Nondegenerate_ETPA_figure for details on the spectrum data
2. In the SPA_ETAP_absorption.m code
   - Change the filenames in line 1-5 to match the filenames of the spectrum data that you are using
     - OPA represents the SPA data
     - The v at the end of the variable indicates that it is for vacuum data
     - Do not worry about the PBE at the end of the variable name.
   - Lines 7-32 generate the two panel SPA and degenerate ETPA figure
     - Major Concerns
       - Line 12 - sets axis for SPA figure
         - Mainly want to make sure that it is the same as for ETPA figure (Line 26)
       - Line 14 - sets position of legend
         - This will need to be manually adjusted if the axis values change or if the size of the figure is changed
       - Line 16 - sets x-axis tics for SPA figure
         - Will need to be modified if x-axis if changed
       - Line 21 - puts (a) in panel
         - Honestly might be simpler to just comment out this line (% is for comments in Matlab)
         - Will also need to be manually adjusted if axis values or figure size change
       - Line 24-25 - plots ETPA data
         - Need to manually set the range to remove tail of ETPA data that correlates to idler energy coming into resonance with SPA
         - This is done by adjust the range of the first dimension that is being plotted for the data
           + e.g. In the code as provided, it is plotted the 1-2301 data point (1:2301)
       - Line 26 - sets axis for ETPA figure

        - Want to allign this with the energy range that you are plotting based on modifications in line 25-26
    - Line 31 - Adds the 2 x ω_i to x-axis label
        - Will also need to be manually adjusted if axis values or figure size change
    - Line 32 - puts (b) in panel
        - Honestly might be simpler to just comment out this line (% is for comments in Matlab)
        - Will also need to be manually adjusted if axis values or figure size change

- Lines 35-71 generate the single panel SPA and degenerate ETPA figure
  - Major Concerns
    - Line 43 - sets axis for figure
        - Make sure that it alligns with data
    - Line 50 - sets info for second axis
        - I set this up to get the desired legend but it makes modifying this figure much more annoying
        - The position vector at the end of this line needs to match the Position (possibly InnerPosition) of the first axis. Unfortunatly this is not the easiest thing to figure out. Assuming that you do not change the size of the figure (actually size of figure not axis range) this value should be fine.
            + Note that I am less confident about this statement if you are doing this on Windows than Mac or Linux, as it forces certain limits on the figure size but what I have should be within those
    - Line 55 - Plots ETPA data
        - Need to manually set the range to remove tail of ETPA data that correlates to idler energy coming into resonance with SPA
        - This is done by adjust the range of the first dimension that is being plotted for the data
            + e.g. In the code as provided, it is plotted the 1-2301 data point (1:2301)
    - Line 65 - puts R6G (in provided code) in figure
        - Will also need to be manually adjusted if axis values or figure size change
    - Line 68-69 - puts legend in figure
        - Position will also need to be manually adjusted if axis values or figure size change and when using different data